

Plenus — Design Implementation Guide (Unity / uGUI)

This guide turns the **Plenus mobile UI design** into your Unity game. It's written for your actual project: Unity 6, Netcode for GameObjects, classic **uGUI** (Canvas + Image + TextMeshPro + LayoutGroups), with your existing `BoardManager` , `DiceManager` , `ScoreUI` , `NetworkMenuManager` , `GameManager` .

0. First: what "the assets" actually are

There are **no PNG/sprite art files to export** from this design, because the design doesn't use any. Every visual is built from three things:

1. **Colors** — solid fills written as `oklch(...)` . I've converted all 52 of them to hex/ `Color` .
2. **Two free Google fonts** — Baloo 2 + Nunito.
3. **CSS shapes** — rounded rectangles, circles, pills, and a few emoji/Unicode glyphs (`★ < > ? * © ×`). No bitmaps.

So "getting all the assets" = this kit. It contains everything the design is actually made of:

```
PlenusDesignKit/
├ Plenus_Implementation_Guide.md      (this file)
├ Scripts/
│   ├── PlenusColors.cs               all 52 colors as UnityEngine.Color
│   ├── ScreenManager.cs              screen switching (the design's nav logic)
│   └ PlenusTheme.cs                  light/dark theme swap + ThemedGraphic
├ Sprites/
│   ├── RoundedRect96_r24.png         9-slice card/button background (white, tint it)
│   ├── Pill160_r32.png               9-slice pill/tag/toggle-track background
│   └ Circle128.png                   avatars, icon buttons, toggle knobs
├ Fonts/
│   └ README_FONTS.md                 where to get the fonts + how to make TMP assets
└ Plenus Board UI (interactive).html  the clickable source design (open in a browser)
```

1. Import the kit into Unity

1. Copy the `Scripts/` files into `Assets/Scripts/UI/` (any folder under Assets).
2. Copy the three PNGs into `Assets/UI/Sprites/` . Select each in the Project window and in the Inspector set **Texture Type = Sprite (2D and UI)**. For `RoundedRect96_r24` and `Pill160_r32` click **Sprite Editor** and set the **9-slice border** to the corner radius:
3. `RoundedRect96_r24` → border **24** on all four sides.
4. `Pill160_r32` → border **32** left/right (top/bottom can stay 0 or 32).
5. `Circle128` → no border (leave as is; only used at fixed sizes).
6. Add the fonts as described in `Fonts/README_FONTS.md` .

That's the whole import. The sprites are pure white so you tint them any color from `PlenusColors` at runtime or in the Inspector.

2. Design tokens (the numbers to reproduce)

2.1 Colors

Light theme (default background chrome)

Token	Hex	Use
LightBg	#F6EDE0	phone background
LightCard	#F9F4EE	cards, panels
LightCardAlt	#E7DCD0	secondary buttons, inputs, chips
LightTextPrimary	#362C24	headings, main text
LightTextSecondary	#6C6158	captions, labels
LightBorder	#E0D6CA	1.5px inset card outlines

Dark theme

Token	Hex
DarkBg	#201914
DarkCard	#2F2722
DarkCardAlt	#3B342E
DarkTextPrimary	#EFEBE4
DarkTextSecondary	#ABA39B
DarkInputBg	#342C26
DarkBorder	#4E463F

Accents (same in both themes)

Token	Hex	Use
PrimaryRed	#D05F43	main call-to-action buttons
PrimaryRedPress	#9D381F	the 3px drop-shadow under red buttons
Green	#418E47	Start Match / Save / positive
GreenPress	#1E6626	drop-shadow under green buttons
ReadyGreen	#38853E	"Ready" status text
AccentLink	#833F27	links ("Forgot password?", "Create account")
LogoutRed	#C74C3D	log-out text
StarGold	#EAB532	stars, gold rank

Board / match colors (the 5 game colors + black wildcard)

Token	Hex
BoardGreen	#80CD82
BoardRed	#F47B74
BoardYellow	#F1D35D
BoardBlue	#79B0E8

Token	Hex
BoardOrange	#FBA962
BoardBlack	#505869 (black wildcard)
CellMarked	#DBD7D0 (a cell after it's marked — greyed)
PendingRing	#0072D5 (blue ring on a temp-selected cell)
DiePip	#362C24 (dots on numeric dice)

These map to your existing `ColorPalette.cs`. The cell inset border in the design is `rgba(40,30,20,0.55)` → `Color(0.157,0.118,0.078,0.55)`.

Avatar swatches (12): ``#CB764E #47944C #488ACB #C99500 #C74B47 #7B63A3 #E1707C #009E98`

B1A93A #887769 #CD6AAF #5D646F`

Rank tiers (low → high): Wood III `#8C6D58`, Wood II `#9E7B64`, Wood I `#B0896F`, Silver II `#ABB2BB`, Silver I `#C2C8CF`, Gold `#E3AE28`, Diamond `#1EBDE3`, Aurelune `#BD5ED2`.

All of the above already live in `PlenusColors.cs`, e.g. `image.color = PlenusColors.PrimaryRed;`

2.2 Typography

- **Baloo 2 ExtraBold** — the "Plenus" logo (40px login / 30px menu), screen titles (19px), button labels (14–17px), big numbers/stats.
- **Nunito Bold/SemiBold** — everything else (11–15px). Uppercase section labels are Nunito ExtraBold 11px, letter-spacing ~0.4px.

2.3 Spacing, radius, shadow

- Phone frame: **390 × 844**, frame radius 44px, screen padding ~22–28px.
- Cards / buttons radius **14–18px** → use `RoundedRect96_r24` 9-sliced (looks right at those sizes).
- Pills / tags / toggle tracks radius **100px** → use `Pill1160_r32`.
- Gaps between stacked elements: **10–18px** (that's your `LayoutGroup Spacing`).
- The signature **chunky button**: a coloured rounded rect with a *darker copy offset 3px down behind it* (`box-shadow: 0 3px 0 <press-color>`). Reproduce with two Images (see §3.2).
- Soft card shadow is subtle; uGUI's `Shadow` component (offset y=3, low alpha) is close enough, or skip it — the design reads fine flat.

3. Build 3 reusable prefabs first

Everything in the design is assembled from these. Build them once, then reuse.

3.1 Card

Image (sprite = `RoundedRect96_r24`, Image Type = **Sliced**), color = `LightCard`. For the "1.5px inset border" look, add a child Image (same sliced sprite) set to `LightBorder`, slightly larger, behind — or just skip the border; it's cosmetic. Add a **VerticalLayoutGroup** + **ContentSizeFitter** if the card should hug its content.

3.2 Button (chunky style)

```

Button (RectTransform)
├ Shadow (Image, sliced RoundedRectangle, color = PrimaryRedPress, offset +3px down, behind)
└ Face (Image, sliced RoundedRectangle, color = PrimaryRed) ← add the Button component here
  └ Label (TMP, Baloo 2 ExtraBold, white)

```

Secondary buttons: Face = `LightCardAlt`, Label = `LightTextPrimary`, and drop the shadow child. Green buttons: Face = `Green`, Shadow = `GreenPress`.

3.3 Toggle (the settings switches)

```

Track (Image, sliced Pill160_r32, size 42×24, color = LightCardAlt when off / Green when on)
└ Knob (Image, Circle128, size 18×18, anchored left when off / right when on)

```

Drive on/off + knob position from a small script hooked to your settings.

Other atoms: **icon button** = Circle128 Image (color `CardAlt`) + TMP glyph (`<`, `?`, `⌂`, `×`). **Tag/chip** = Pill160_r32 Image + TMP. **Avatar** = Circle128 Image tinted, + TMP initial/emoji.

4. How the design's concepts map to uGUI

Design (HTML/CSS)	uGUI equivalent
<code><sc-if nav==='menu'></code> (one screen)	one panel GameObject ; <code>ScreenManager.Show()</code> enables one at a time
<code>display:flex; flex-direction:column; gap:14px</code>	VerticalLayoutGroup , Spacing = 14
<code>display:flex; gap (row)</code>	HorizontalLayoutGroup
<code>flex:1</code> spacer (pushes to bottom)	empty object with LayoutElement → Flexible Height = 1
<code><sc-for list=...></code> (repeated rows)	instantiate a row prefab in a C# loop
<code>{{ nav2.profileName }}</code> binding	<code>text.text = ...</code> in C#
<code>justify-content:space-between</code>	HorizontalLayoutGroup + a flexible spacer, or anchor children
<code>grid-template-columns:repeat(2,1fr)</code>	GridLayoutGroup (2 columns) — you already use this for the board
<code>border-radius</code>	sliced rounded sprite (§1)
<code>position:absolute; inset:0</code> overlay	full-stretch panel on top (modals: Rank info, settings)
phone 390×844	CanvasScaler : Scale With Screen Size, ref 390×844 (or 1080×2340), Match ≈ 0.5

Screens present in the design (build panels for each): Login · Main Menu · Configuration · Ranked (+ "how ranking works" modal) · Create Lobby · Join Lobby · Profile · Personalize Avatar · Scoring Rules · Initiative roll · Match (board) · Opponent-board view · live-feedback toast.

5. Screen-by-screen

Each screen is one panel under `ScreenRoot`, background = theme `PhoneBg`, padding via a VerticalLayoutGroup with left/right padding ~24. Only the specifics differ:

Login — logo "Plenus" (Baloo 40) + "Board Duel" caption; two inputs (`TMP_InputField` , sliced rounded, `InputBg`); right-aligned "Forgot password?" (AccentLink); red **Log in** button; an "or" divider (two thin `Border` images + label); secondary **Continue as guest**; footer "New here? Create an account". Theme toggle icon top-right. `Log in` → `ShowMenu()` .

Main Menu — top-left gear icon (→ Config), top-right `?` (→ Rules) and `⌂` (theme). Logo. A profile card (avatar + name + rank chip + `>`) → `ShowProfile()` . Then three big buttons: **Create Lobby** (red) → CreateLobby, **Join Lobby** (cardAlt) → JoinLobby, **Ranked** (cardAlt) → Ranked. Bottom link "Resume match in progress" → Match.

Configuration — back `<` , title. Sections (uppercase labels): *Appearance* = Light/Dark segmented control (two pills, active = card, calls `PlenusTheme.SetDark()`). *Sound & haptics* = a card with 4 toggle rows (Sound effects, Music, Vibration, Push notifications) using the §3.3 toggle. *Account* card. Red **Log out** text. Version footer.

Ranked — back + title + `i` (opens modal). A big tier card: rotated-45° rounded square in the tier color + "Overall Rating" + tier label + "Global rank #N · x/10 wins". A progress row of 10 pips (filled = `Green` /tier, empty = `CardAlt`). "Leaderboard" list via row prefab (rank #, tier diamond, name + tier label, wins). Row for "you" is highlighted. Modal "How ranking works": tier table (8 rows: color diamond, label, "top X% of players").

Create Lobby — back + title. "Lobby code" big centered pill (`PLN-482` , letter-spacing 4px). "Players in lobby" list (avatar + name + green "Ready"). Bottom green **Start Match** → `ShowInitiative()` .

Join Lobby — back + title. "Enter lobby code" input. "Open lobbies nearby" list (avatar + name + "1/2 players" + red **Join** pill → Initiative).

Profile — back + title. Big 96px avatar (tinted circle + initial). Name (Baloo 20). 2-column **GridLayoutGroup** of stat tiles (Matches 24, Wins 15, Win rate 63% [green], Win streak 3 [red], Best streak 6, Stars 128, Rows completed 41, Colours completed 9) — each tile is a `Card` with a big Baloo number + Nunito caption. "Your nemesis" card. Bottom **Personalize avatar** → Avatar.

Personalize Avatar — back + title. 110px live-preview avatar. "Choose an icon" = wrap of 10 rounded-square emoji buttons (`🦊🐻🐼🐱🐶🐷🐸🐙🦉🐰`), selected one gets a ring (`box-shadow` → a highlighted outline Image). "Choose a colour" = wrap of 12 circle swatches (the 12 avatar colors), selected gets a ring. Bottom green **Save** → Profile.

Scoring Rules — back + title, scrollable (**ScrollRect** + `Mask` + `VerticalLayoutGroup` content). "Row bonuses" = 15 rows (A–O), each a mini bar + "1st +x" (gold) / "2nd +y". Then "Colour bonus", stars penalty, jokers, etc. Values in §8. Back returns to whoever opened it (`ScreenManager.Back()`).

Initiative — both players roll to see who starts; a rolling animation then a result. Wire to your real initiative logic; visually it's two dice + names + a "You go first" banner.

Match (the board) — **you already have this**. Reskin, don't rebuild: - Top bar: two player chips ("M You" red, "A Alex" blue) + `i` + settings circle. - Row of joker diamonds + "jokers" + "Your board" + `★ 0/15` . - The **15×7 board**: your `BoardManager` `GridLayoutGroup`. Cell = Image (sliced, radius 7, size ~33) tinted `PlenusColors.Board(colorName)` , inset border, `★` overlay for star cells, greyed to `CellMarked` when committed, blue `PendingRing` when temp-selected. Row status tags ("1st +1" gold / "2nd +1") sit to the right of the relevant rows. - Bottom "felt" panel: current dice (3 color squares + 3 numeric dice with pips), "7 jokers left — use one to ignore:" with **Skip colour** / **Skip number** pills, then **Roll** (red) / **Confirm** (grey/green) / **Pass** buttons. These map to your `DiceManager` / `DiceButtonUI` / `SelectionManager` and the Confirm/Pass/Roll buttons already in your Hierarchy.

Opponent view + toast — a toggle flips the board to the opponent's; a small toast ("Alex just completed row H!") slides in for ~2.8s on opponent moves. Toast = a Pill/Card at the top, shown/hidden by a coroutine.

6. Wiring navigation & theme

- Add `ScreenManager` to `ScreenRoot`, register every panel with its `Screen` enum value, set `startScreen = Login`. Each nav button's `OnClick` calls the matching `Show...()` method (e.g. Create Lobby button → `ShowCreateLobby`).
- For light/dark: put `ThemedGraphic` on every Image/TMP that should recolor, set its `Role` (`PhoneBg`, `Card`, `TextPrimary`, ...). The theme toggle button calls `PlenusTheme.Toggle()`. Accent colors (red/green/gold) are theme-independent — leave those as fixed `PlenusColors`.
- The design's screen flow: Login → Menu → (Create/Join Lobby) → Initiative → Match, with Menu also reaching Config, Ranked, Profile → Avatar, and Rules from the `?`. Hook the lobby/match transitions to your real `NetworkMenuManager` (LobbyScene → MainGameScene via `NetworkSceneManager`) rather than just toggling panels once networking is involved.

7. Map to your existing scripts

Design piece	Your project
Match board rendering	<code>BoardManager</code> , <code>CellView</code> , <code>GridUtils</code> , <code>BoardGenerator</code> — reskin cell visuals with <code>PlenusColors.Board()</code>
Dice panel + selection	<code>DiceManager</code> , <code>DiceButtonUI</code> , <code>DiceButton</code> , <code>SelectionManager</code>
Score / stats / bonuses	<code>ScoreManager</code> , <code>ScoreUI</code> , <code>ScoreConfig</code> , <code>ScoreBreakdown</code> , <code>PlayerData</code> (Profile & Rules screens read from these)
Lobby / menu flow	<code>NetworkMenuManager</code> , <code>GameManager</code> — drive <code>ScreenManager</code> from these
Colours	fold <code>PlenusColors</code> into your <code>ColorPalette.cs</code> (or reference it directly)

Reskin first (fastest win): the Match screen — new colors, fonts, rounded cells, chunky buttons — no new logic. **Then build new screens:** Menu is the best second step because it establishes your Card/Button prefabs; every other screen reuses them.

8. Data the design encodes (reference)

Your gameplay already implements most of this; included so the UI shows the right numbers.

- **Board:** 15 rows (A–O) × 7 cols = 105 cells. 5 colors, each 21 cells. 2 stars per color (STAR total shown as 15 in the design's counter). Black = colour wildcard, numeric **6** = number wildcard.
- **Marking rule (from the design's logic):** with numeric die value N , a cell is markable if `row+1 == N` or `col+1 == N` (plus colour match, unless a wildcard/joker is used).
- **Jokers:** max **8** per game; "Skip colour"/"Skip number" each spend one.
- **Row rewards** {1st, 2nd} for rows A→O: (5,3)(3,2)(3,2)(3,2)(2,1)(2,1)(2,1)(1,0)(2,1)(2,1)(2,1)(3,2)(3,2)(3,2)(5,3) — the 1st-place column matches your `context.md` reward array [5,3,3,3,2,2,2,1,2,2,2,3,3,3,5].
- **Colour bonus:** first to finish a colour +5, second +3.
- **Penalty:** -2 per unmarked star. **Unused joker:** +1 each.
- **Rank tiers:** Wood III/II/I, Silver II/I, Gold, Diamond, Aurelune; 10 wins to advance a tier.
- **Avatars:** 10 emoji icons + 12 colours (listed in §2.1).

9. Suggested order (checklist)

1. Import kit (fonts, sprites as 9-slice, scripts). Set CanvasScaler to 390×844.
2. Build the 3 prefabs: Card, Button, Toggle (+ icon button, tag, avatar atoms).
3. Reskin the **Match** screen with the new palette/fonts/cells/buttons.
4. Build **Main Menu** (locks in your prefabs + `ScreenManager`).
5. Build Login, Config (+ theme toggle), Profile, Avatar.
6. Build Create/Join Lobby + Ranked + Rules + Initiative.
7. Wire lobby/match transitions into `NetworkMenuManager`; hook Profile/Rules to `PlayerData` / `ScoreManager`.
8. Add light/dark `ThemedGraphic` pass + the opponent-view toggle and toast.

Open `Plenus Board UI (interactive).html` in a desktop browser side-by-side while you build — it's the living spec for spacing, states, and interactions.